Rasel Ahmed Bhuiyan

Computer Vision & Machine Learning · Foundation Models · Generative AI · Biomedical Imaging

Mishawaka, IN | (574) 993-4750 | rbhuiyan@nd.edu | bhuiyanra.github.io | linkedin.com/in/rabhuiyan

SUMMARY

Machine learning researcher who builds and evaluates computer vision models for biometric and biomedical images when labeled data is scarce and shifts over time, across sites, or across populations. Adapts vision foundation models (DINOv3) with metric learning, LoRA, and multiple instance learning, builds generative and vision-language models, and trains on multi-GPU clusters. Built a canthi detection model that runs in NIST's internal iris recognition pipeline. 14 peer-reviewed publications; Best Paper Award, WACV 2025 (CV4Smalls).

TECHNICAL SKILLS

Machine Learning & Vision: CNNs, vision transformers, foundation-model adaptation (frozen, partial unfreezing, LoRA), metric learning (ArcFace, triplet), attention-based MIL, multi-task learning, segmentation (U-Net), landmark localization, image regression, NIR+RGB fusion, saliency-guided training

Generative & Multimodal AI: GANs (StyleGAN2-ADA), diffusion models, synthetic data generation, image–text models (SigLIP, BioCLIP), multimodal LLMs (LLaMA-3.2-Vision, Qwen-3-VL), retrieval-augmented prompting

Evaluation: Subject-disjoint and cross-site validation, cross-dataset transfer, demographic bias analysis, EER/ROC analysis, ISO/IEC 29794-6 image quality

Languages & Tools: Python, PyTorch, TensorFlow, Hugging Face, NumPy, Pandas, scikit-learn, OpenCV, ONNX, W&B, Bash, SQL, MATLAB, Git, Docker, Linux, CUDA, HPC clusters (SGE, SLURM), multi-GPU training

RESEARCH EXPERIENCE

Graduate Research Assistant — Jan. 2022 – Present

Computer Vision Research Laboratory (CVRL), University of Notre Dame — *Notre Dame, IN*

- **ColdIrisNet** — Foundation-Model Adaptation under Distribution Shift (under review, Information Fusion)
  - Adapted a frozen DINOv3 (ViT-L/16) with ArcFace metric learning and a saliency-guided multi-task loss that keeps the model on iris texture rather than decomposition artifacts.
  - Designed a landmark-localization network that keeps features aligned under rotation of up to $\pm 80°$.
  - Cut Equal Error Rate (EER) by 68% vs. prior state of the art (EER $\approx$ 2.9%) across 70 days of post-mortem change; showed strong cross-dataset and live-to-deceased transfer.
- **Post-Mortem Interval (PMI) Estimation** — Multimodal Image Regression (WACV 2025; ongoing)
  - Developed the first method to estimate time since death from iris images; an NIR+RGB fusion model reached 3.4-hour MAE on same-distribution data and 17.5 hours on held-out subjects. [code]
  - Cut cross-laboratory error (U.S. and Poland data) by 40% by balancing training data with synthetic images.
  - Adapted DINOv3 (frozen, partial unfreezing, LoRA) with attention-based MIL (ABMIL): 31.4% lower MAE than state-of-the-art iris-based PMI estimators (paper in preparation).
  - Built subject-disjoint data split tools so reported errors reflect performance on new subjects.
- **Multimodal LLMs & Vision-Language Models** — Estimation with Limited Data
  - Built a retrieval-augmented few-shot pipeline: multimodal LLMs (LLaMA-3.2-Vision, Qwen-3-VL) estimate PMI from MPNet-retrieved examples; benchmarked vs. ResNet-152 and foundation-model baselines.
  - *In progress:* training SigLIP- and BioCLIP-based image–text models that predict PMI from an NIR image and its expert description (frozen encoders, last-layer fine-tuning, LoRA).
- **Infant Iris Recognition** — Clinical Image Analysis (NSF-funded; Best Paper Award)
  - Built the first operational infant iris recognition system, evaluated on a clinical population (EER $\approx$ 3%, AUC 0.99); some commercial systems designed for adults rejected every image in the dataset.
  - Trained a Nested Residual U-Net with no infant labels, using infant-specific augmentations of adult images.
  - Released a privacy-preserving synthetic infant iris dataset. [code]

- **Iris Presentation Attack Detection** — Large-Scale Training
  - Trained models on 400K+ images of live eyes and six attack types (printouts, synthetic irises, Kindle displays, textured contacts, post-mortem and artificial eyes); over 95% accuracy on unseen attack types.
  - Built the training pipeline with crop caching and multi-GPU training.
- **Synthetic Data** — Conditional Generative Modeling (WACV 2024 Workshop)
  - Built a PMI-conditioned StyleGAN2-ADA generator with latent hypersphere sampling for controllable identities; generated 180K images across 18 PMI ranges (11/12 ISO/IEC 29794-6 checks passed). [code]
- **Iris Image Processing Toolbox** — Deployed at NIST
  - Delivered models for iris boundary estimation, eye-canthi localization and alignment, eyelid curve fitting, and projective gaze correction to the NIST IREX-vetted OpenSourceIrisRecognition toolkit. [code]
  - Canthi detection model runs in NIST's internal iris recognition pipeline.
- **Large-Scale Evaluation & Fairness** — Post-Mortem Iris Data (IEEE T-BIOM 2025)
  - Analyzed 5,770 NIR and 4,643 visible-light images from 259 subjects; ran a demographic bias analysis across 32M+ comparisons covering age, gender, and decomposition stage.
- **Perceptographer** — Generative AI for Neuroscience (NIMH collaboration)
  - Designed a hybrid GAN–diffusion engine that visualizes brain-stimulation-induced perceptual events.

Lecturer and Research Assistant 2016 – 2021
Uttara University & University of Asia Pacific *Dhaka, Bangladesh*

- Built ML and signal processing pipelines for EEG seizure detection, ECG arrhythmia classification, and wearable-sensor activity recognition (5 publications).
- Built sign language and gesture recognition using CNN features and transfer learning (3 publications).
- Lecturer, Uttara University (2018–2021): taught Algorithms, Digital Logic Design, Machine Learning, and AI.

EDUCATION

Ph.D., Computer Science & Engineering, University of Notre Dame, IN — Expected by May 2027
M.Sc., Computer Science & Engineering, University of Notre Dame, IN — Aug. 2025
B.Sc., Computer Science & Engineering, University of Asia Pacific, Dhaka, Bangladesh — Mar. 2018

SELECTED PUBLICATIONS

14 peer-reviewed publications. Full list on Google Scholar.

[1] **R.A. Bhuiyan**, et al., "Iris Recognition for Infants," *WACV Workshops*, 2025. **Best Paper Award**

[2] **R.A. Bhuiyan**, A. Czajka, "Forensic Iris Image-Based Post-Mortem Interval Estimation," *WACV*, 2025.

[3] **R.A. Bhuiyan**, et al., "Beyond Mortality: Advancements in Post-Mortem Iris Recognition through Data Collection and Computer-Aided Forensic Examination," *IEEE T-BIOM*, 2025.

[4] **R.A. Bhuiyan**, A. Czajka, "ColdIrisNet: Biology-Guided Fusion for Forensic Iris Recognition," *Information Fusion*, under review.

[5] **R.A. Bhuiyan**, A. Czajka, "Forensic Iris Image Synthesis," *WACV Workshops*, 2024.

[6] E. Shahbazi, D. Nguyen, **R.A. Bhuiyan**, A. Czajka, A. Afraz, "Visualizing the Unseen: Perceptographer, an AI Engine for Visualizing Brain-Stimulation-Induced Perceptual Events," *Journal of Vision*, 2025 (abstract).

HONORS, SERVICE & COLLABORATIONS

- Best Paper Award, IEEE/CVF WACV 2025 CV4Smalls Workshop | NSF-supported research (Grant 2237880)
- Invited talk, Voxel51 (2025) | Reviewer, IEEE/CVF WACV 2025 and 2027 | Mentored two undergraduates
- Collaborations: NIST, NIMH, Michigan State University, Medical University of Warsaw, and Dutchess County Medical Examiner's Office